

this, as they can prototype their whole chip on this FPGA. The type of applications this would serve are 400G optical transport network (OTN) switching, 400G transponder (used by back-end metro networks), 2x100G Muxponder and ASIC prototyping or prototyping of chip using FPGA.

In the medical field, today's prevalent ultrasound technology allows only 128 channels. But this new technology allows 256 channels. It has much higher resolution and the vision is clearer. Finally, in transmit and receive data processing, we will be able to support 48 channels. The key thing is, for these kind of applications, we were using two non-3D IC-based FPGAs, but now with just one FPGA with a 3D IC, we will be able to do the same.

There are a lot of process nodes available here. How to select one?

We have been building FPGAs at different nodes – from 130nm to 90nm to 45/40nm. In the past, when we had 130nm and introduced 90nm, the 90nm basically replaced 130nm, and then 45/40nm replaced 90nm. But now due to technology and business-related reasons, we will have concurrent nodes as portfolio. That means we will have 28, 20 and 16nm coexisting. None will replace each other because each one is solving a different challenge, addressing a different market or application requirement. So 28nm will have a long life, and this family is indeed the leader in price performance per watt. The 20nm announcement we

made recently complements 28nm bandwidth where we have high-performance architecture. We also announced our plans of 16nm, which complements 20nm.

What technology enables you to build an FPGA on the 16nm node?

That is how we are able to bring more value. The new technology used for 16nm is called FinFET. Till about 130nm, as you move around the process nodes, the power and cost reduces. Beyond that, there is a new phenomenon that came up. As transistors become small, they start leaking. Power (especially static power) goes up because of this. At 28nm and 20nm, this was the big problem for the industry. We tried to solve this through architecture, choosing the right process. But then the fabs came up with FinFET technology that solves this problem.

What is the most exciting area that FPGAs power today?

The 28nm node is enabling programmable systems. I think the commonly used term here is 'smarter.' Earlier, if you had bandwidth you could transfer data. Now what we need more is to transmit 'intelligent' data. Let us take the case of smarter vision, which cuts across different market segments. Image or video-related applications could be automotive, 3D surrounding, medical imaging or industrial machines. An example for machine vision is a rice sorter that sorts rice grains based on the size of the grains. The machine looks at the rice and sorts it out. Those are part of our smarter vision campaign. ●

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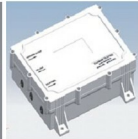
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Rajamane Telectric Pvt. Ltd.

#3, Doddanakundi Indl Area, Mahadevapura, Bangalore - 560048, INDIA. Tel: +91 -80-42483700

Fax: +91-80-42483777 E-mail: salessolar@rajamane.in, sales@rajamane.in www.rajamane.in